

SEPTEMBER 2010

Teacher: _____

[illegible]

Question 1**Marks**

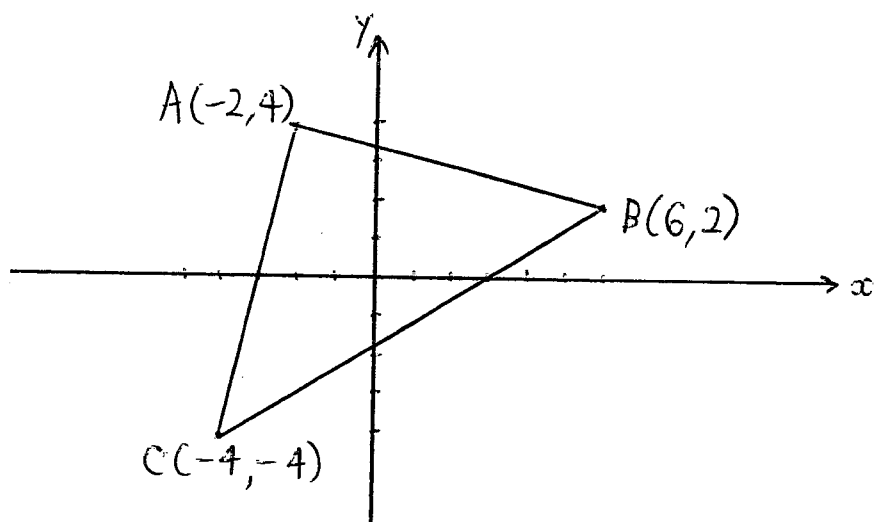
- a) Simplify $\frac{a^4(b^2)^5}{a^3b^4}$ 2
- b) Evaluate $\frac{15.73-8.27}{\sqrt[3]{2.43}}$ correct to 3 significant figures. 2
- c) Find the values of a and b if $\frac{2}{2+\sqrt{3}} = a - \sqrt{b}$ 2
- d) If $\tan\theta = \frac{-8}{15}$, find $\sin\theta$, if $90 < \theta < 180$ (leave in fraction form) 2
- e) Simplify $\frac{9x^2-4}{6x^2-x-2}$ 2

Question 2 (Start on a new page)

- a) Solve $|3x + 2| \geq 8$ 2
- b) i) Sketch the graph $y = |x - 1|$ 1
State its
ii) domain 1
iii) range 1
- c) Solve $x^6 + 7x^3 - 8 = 0$ using the substitution $V = x^3$ 3
- d) Find $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2}$ 2

Question 3 (Start on a new page)**Marks**

- a) The diagram below shows the points $A(-2,4)$, $B(6,2)$ and $C(-4,-4)$



- | | | |
|------|--|---|
| i) | Calculate the length of the interval BC | 1 |
| ii) | Find the gradient of the line BC . | 1 |
| iii) | Find the coordinates of M , the midpoint of BC . | 1 |
| iv) | Show that the equation of l , the perpendicular bisector of BC is $5x + 3y - 2 = 0$. | 2 |
| v) | Show that l passes through A . | 1 |
| vi) | Given the equation of BC is $3x - 5y - 8 = 0$ find the perpendicular distance of A from BC | 2 |
| vii) | Hence or otherwise find the area of $\triangle ABC$. | 1 |
- b) Find the angle sum of a 15 sided polygon. 1

Question 4 (Start on a new page)

- | | | |
|------|---------------------|---|
| a) | Differentiate | |
| i) | $-3x^4$ | 1 |
| ii) | $(4x - 9)^5$ | 1 |
| iii) | $4x^2(6x-5)^5$ | 2 |
| iv) | $\frac{4x-3}{3x+4}$ | 2 |
- b) i) Find the point on the curve $y = x^2 + 2$ where the tangent has a gradient of -4 . 2
- ii) Find the equation of the normal at this point and give your answer in general form. 2

Question 5 (Start on a new page)**Marks**

- a) Solve $x^2 - 7x + 12 > 0$ and sketch your solution on a number line. 3
- b) Solve $|2x + 6| = 3x - 1$ and show that only one solution is valid. 3
- c) i) Find the discriminant of $2x^2 - 3x + K$ 1
- ii) For what values of K is the expression $2x^2 - 3x + K$ positive definite? 2
- iii) For what value of K does $2x^2 - 3x + K$ have equal roots? 1

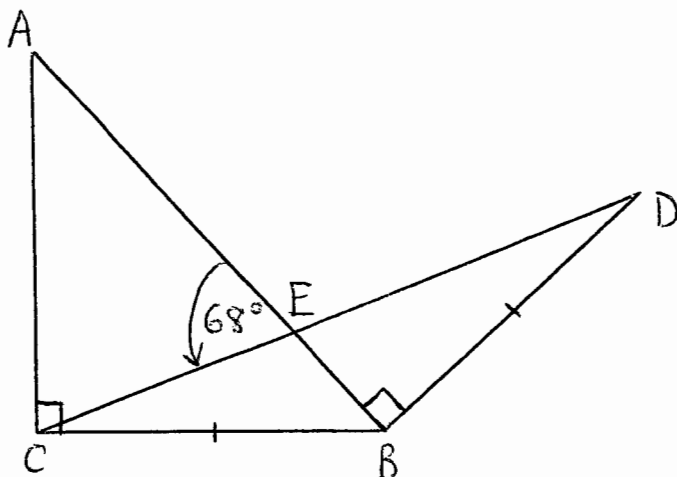
Question 6 (Start on a new page)

- a) Prove that $\frac{1}{1+\cot^2\theta} = (1 - \cos\theta)(1 + \cos\theta)$ 3
- b) Find the values of A, B and C if $3x^2 - 7x + 5 \equiv Ax(x - 1) + Bx + C$ 3
- c) If α and β are the roots of the equation $x^2 - 2x - 7 = 0$ find the value of
- i) $\alpha + \beta$ 1
- ii) $\alpha\beta$ 1
- iii) $\alpha^3\beta + \alpha\beta^3$ 2

Question 7 (Start on a new page)

- a) If $f(x) = (x + 1)(x + 2)$ find in simplest terms an expression for $f(x + 1) - f(x)$. 2

- b) 3



Copy the above diagram into your answer booklet.

In the diagram, above find $\angle DCB$ giving reasons.

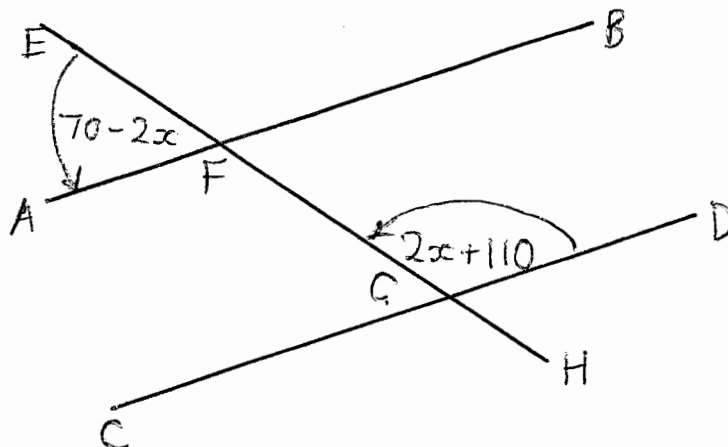
- c) i) Sketch $y = x + 1$ and $y = x - 2$ on the same graph. 1
- ii) Shade the region $y \geq x + 1$ 1
- iii) What inequality describes the region between and including the two lines? 1
- d) Sketch $y = \frac{2}{x-2}$ showing all important features. 2

Question 8 (Start on a new page)

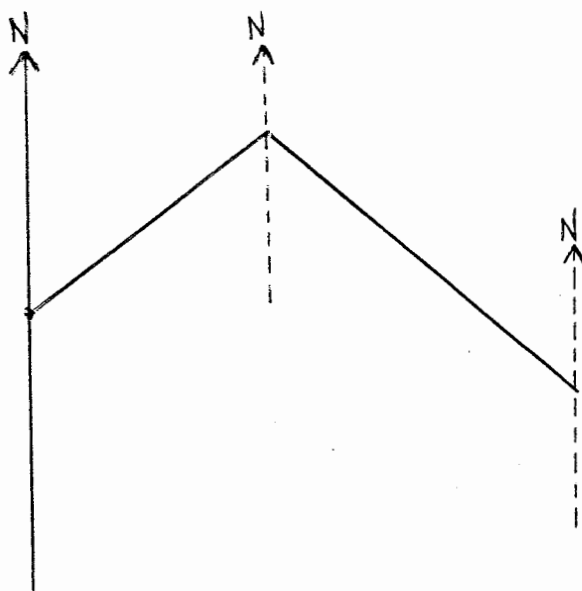
Marks

- a) Prove that AB and CD are parallel.

2



- b)



A ship sails for 270 n. miles from Sydney on a bearing of $055^\circ T$. It then turns and sails for 360 n. miles on a bearing of $120^\circ T$.

- i) Copy the above diagram onto your answer sheet and show the information given.

1

- ii) How far is the ship from Sydney to the nearest nautical mile?

2

- iii) Find the bearing of Sydney from the ship to the nearest degree

2

- c) Solve $\cos^2 \theta = \frac{3}{4}$ for $0^\circ \leq \theta \leq 360^\circ$.

3

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Year 11 Mathematics Yearly 2010 Solutions

Question 1

$$a) \frac{a^4 b^{10}}{a^3 b^4} = ab^6$$

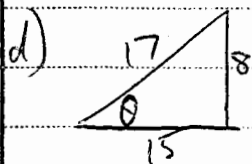
$$b) 5.55$$

$$c) \frac{2}{2+\sqrt{3}} = a - \sqrt{b}$$

$$\frac{2}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{4-2\sqrt{3}}{1} = a - \sqrt{b}$$

$$4 - \sqrt{12} = a - \sqrt{b}$$

$$a = 4 \quad b = 12$$



$$\sin \theta = \frac{8}{17}$$

$$e) \frac{(3x-2)(3x+2)}{(3x-2)(2x+1)} = \frac{3x+2}{2x+1}$$

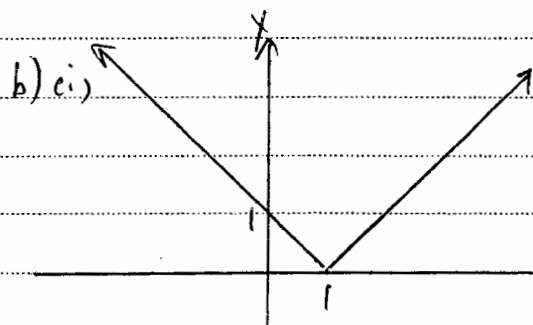
Question 2

$$a) |3x+2| \geq 8$$

$$3x+2 \geq 8 \quad \text{or} \quad 3x+2 \leq -8$$

$$3x \geq 6 \quad \text{or} \quad 3x \leq -10$$

$$x \geq 2 \quad \text{or} \quad x \leq -\frac{10}{3}$$



$$\text{cii) } D: \text{ All real } x$$

$$c) x^6 + 7x^3 - 8 = 0$$

$$V^2 + 7V - 8 = 0 \quad \text{if } V = x^3$$

$$(V-1)(V+8) = 0$$

$$V = 1 \quad \text{or} \quad -8$$

$$x^3 = 1 \quad \text{or} \quad x^3 = -8$$

$$x = 1 \quad \text{or} \quad x = -2$$

$$\text{ciii) } R: y \geq 0$$

$$d) \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

$$\lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)}$$

$$= 4$$

Question 8

$$a) \angle BFG = 70 - 2x \quad (\text{vertically opp.})$$

$$\angle BFG + \angle FGD$$

$$70 - 2x + 2x + 110 = 180$$

$\therefore AB \parallel CD$ (co-interior angles are supplementary)

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$$a) \text{ci) } BC = \sqrt{(6-(-4))^2 + (2-(-4))^2} \quad \text{cii) } M_{BC} = \frac{2-(-4)}{6-(-4)}$$

$$= \sqrt{10^2 + 6^2}$$

$$= \sqrt{136}$$

$$= \frac{6}{10}$$

$$= \frac{3}{5}$$

$$\text{ciij) } M_{BC} : \left(\frac{-4+6}{2}, \frac{-4+2}{2} \right) \quad \text{civ) } m = -\frac{5}{3} \quad (1, -1)$$

$$y - (-1) = -\frac{5}{3}(x - 1)$$

$$3y + 3 = -5x + 5$$

$$5x + 3y - 2 = 0$$

$$= (1, -1)$$

$$\text{cv) } (-2, 4) \text{ satisfies } 5x + 3y - 2 = 0 \quad \text{cvij) } d = \frac{|3 \times -2 + 5 \times 4 - 8|}{\sqrt{3^2 + (-5)^2}}$$

$$\text{if } 5x - 2 + 3 \times 4 - 2 = 0$$

$$-10 + 12 - 2 = 0$$

$$2 - 2 = 0 \checkmark$$

$$= \frac{34}{\sqrt{34}} = \sqrt{34}$$

 $\therefore A$ lies on line.

$$\text{cvij) } A = \frac{1}{2}bh$$

$$= \frac{1}{2} \times \sqrt{136} \times \sqrt{34}$$

$$A = 34 \text{ units}^2$$

$$b) \text{Angle Sum} = (n-2) \times 180$$

$$= 13 \times 180$$

$$= 2340^\circ$$

Question 4

$$a) \text{ci) } -3x^4 \quad \text{cii) } (4x-9)^5 \quad \text{ciij) } 4x^2(6x-5)^5$$

$$\frac{d}{dx} = -12x^3 \quad \frac{d}{dx} = 5(4x-9)^4 \times 4 \quad \frac{d}{dx} = 4x^2 \times 5(6x-5)^4 \times 6$$

$$= 20(4x-9)^4 \quad + 8x(6x-5)^5$$

$$= 120x^2(6x-5)^4 + 8x(6x-5)^5$$

$$\text{civ) } \frac{4x-3}{3x+4}$$

$$\frac{d}{dx} = \frac{(3x+4)4 - (4x-3)3}{(3x+4)^2} = \frac{25}{(3x+4)^2}$$

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b. ci) $y = x^2 + 2$

$\frac{dy}{dx} = 2x$

$-4 = 2x$

at $x = -2$:

Pt. is $(-2, 6)$

cii) $M = \frac{1}{4} (2, 6)$

$y - 6 = \frac{1}{4}(x + 2)$

$4y - 24 = x + 2$

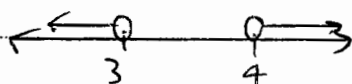
$x - 4y + 26 = 0$

Question 5

a) $x^2 - 7x + 12 > 0$

$(x - 4)(x - 3) > 0$

$x > 4, x < 3$



b) $|2x + 6| = 3x - 1$

$2x + 6 = 3x - 1$ or $2x + 6 = -(3x - 1)$

$7 = x$ or $5x = -5$

$x = -1$

Sub. $x = 7$

Sub. $x = -1$

$|14 + 6| = 21 - 1$ ✓ $|-2 + 6| \neq -4$ ✗

$\therefore$ only $x = 7$ valid

c) ci) $\Delta = b^2 - 4ac$

$= (-3)^2 - 4 \times 2 \times k$

$= 9 - 8k$

cii) If $\Delta < 0$

$9 - 8k < 0$

$k > \frac{9}{8}$

ciii) $\Delta = 0$

when $k = \frac{9}{8}$

Question 6

a) $\frac{1}{1 + \cot^2 \theta} = (1 - \cos \theta)(1 + \cos \theta)$

$\frac{1}{\operatorname{cosec}^2 \theta} = (1 - \cos \theta)(1 + \cos \theta)$

$\sin^2 \theta =$ "

$1 - \cos^2 \theta =$ "

$(1 - \cos \theta)(1 + \cos \theta) = \text{RHS}$

b) $3x^2 - 7x + 5 \equiv Ax(x - 1) + Bx + C$

When $x = 0$

$5 = C$

When $x = 1$,

$1 = B + C \therefore B = -4$

When $x = 2$

$3 = 2A - 8 + 5$

$A = +3$

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$$\begin{aligned} \text{c) (i) } \alpha + \beta &= \frac{-b}{a} \\ &= \frac{2}{1} \\ &= \underline{2} \end{aligned}$$

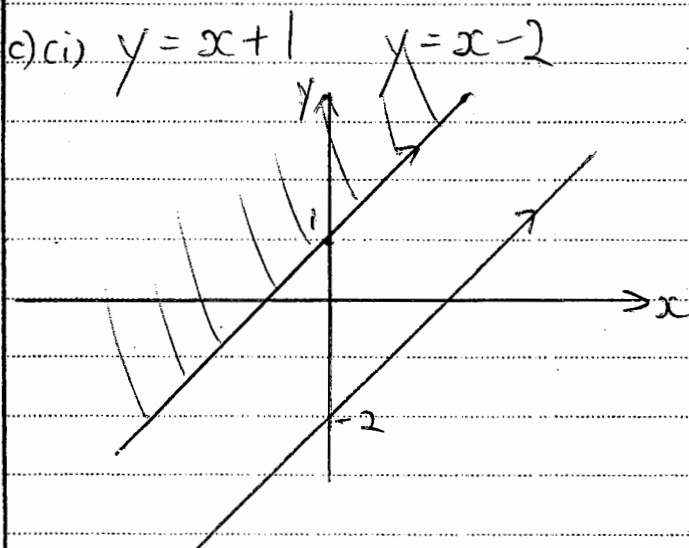
$$\begin{aligned} \text{c) (ii) } \alpha \beta &= \frac{c}{a} \\ &= \frac{-7}{1} \\ &= \underline{-7} \end{aligned}$$

$$\begin{aligned} \text{c) (iii) } \alpha^3 \beta + \alpha \beta^3 &= \alpha \beta (\alpha^2 + \beta^2) \\ &= -7 [(\alpha + \beta)^2 - 2\alpha\beta] \\ &= -7 [2^2 - 2(-7)] \\ &= \underline{-126} \end{aligned}$$

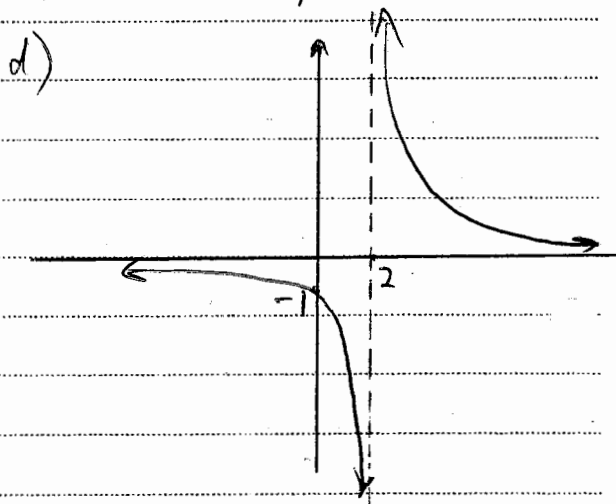
Question 7

$$\begin{aligned} \text{a) } f(x+1) - f(x) &= (x+1+1)(x+1+2) - (x+1)(x+2) \\ &= x^2 + 5x + 6 - x^2 - 3x - 2 \\ &= \underline{2x + 4} \end{aligned}$$

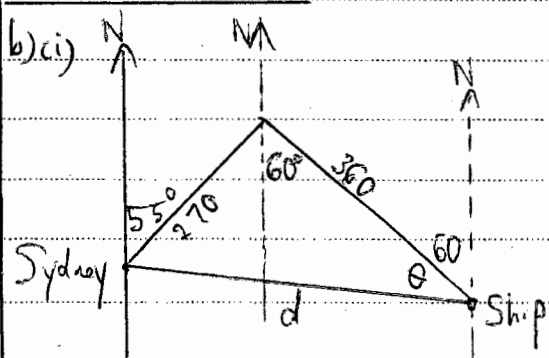
$$\begin{aligned} \text{b) } \angle DEB &= 68^\circ \text{ (vertically opp.)} \\ \angle EDB &= 22^\circ \text{ (< sum of } \Delta) \\ \angle DCB &= \underline{22^\circ} \text{ (equal <'s opp equal sides)} \end{aligned}$$



c) (ii) $x - 2 \leq y \leq x + 1$



Question 8



$$\text{c) (ii) } d^2 = 270^2 + 360^2 - 2 \times 270 \times 360 \cos 115$$

$$\underline{d = 534 \text{ n. miles}}$$

$$\text{c) (iii) } \frac{\sin \theta}{270} = \frac{\sin 115}{534}$$

$$\sin \theta = 0.458$$

$$\theta = 27^\circ \text{ (nearest degree)}$$

$$\begin{aligned} \text{c) } \cos^2 \theta &= \frac{3}{4} \\ \cos \theta &= \pm \frac{\sqrt{3}}{2} \end{aligned}$$

Working θ is 30°

$$\therefore \theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$

$\therefore$ Bearing is 273° T

or $N 87^\circ W$